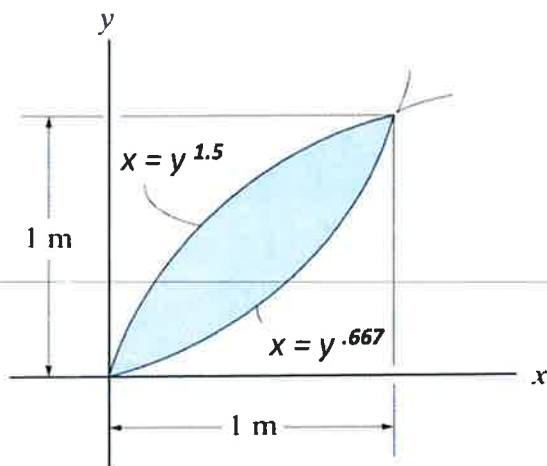


Section 4 – Centroids

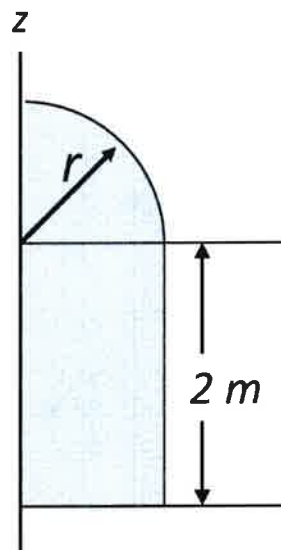
1. (5 pts) When using a **horizontal** differential element of area to solve for the location of $\bar{x}$ for the shaded area, what expression would be correct for dA ?

- A. $(y^{0.667} - y^{1.5}) dy$
- B. $(y^{0.667} + y^{1.5}) dy$
- C. $0.5(y^{1.5} - y^{0.667}) dy$
- D. $(y^{1.5} - y^{0.667}) dy$
- E. $0.5(y^{0.667} + y^{1.5}) dy$



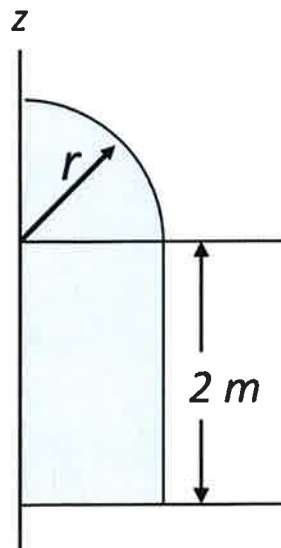
2. (5 pts) Determine the outside **surface area** of the solid formed by revolving the shaded area 360° about the z axis if $r = 1.5$ m. Do not include the bottom surface/floor of the solid.

- A. 28.3 m²
- B. 34.6 m²
- C. 29.9 m²
- D. 32.0 m²
- E. 33.0 m²



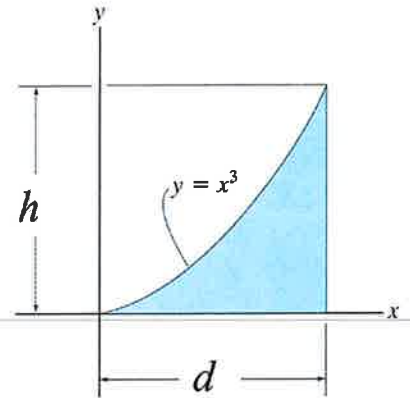
3. (5 pts) Determine the **volume** of the solid formed by revolving the shaded area 360° about the z axis when $r = 1.5$ m.

- A. 20.0 m³
- B. 21.2 m³
- C. 23.2 m³
- D. 18.1 m³
- E. 16.7 m³



4. (5 pts) Determine the location of $\bar{x}$ of the shaded area if $d = 1.5$ m, and the area of the shaded part is 1.266 m^2 .

- A. 1.04 m
- B. 0.96 m
- C. 1.20 m
- D. 0.83 m
- E. 1.14 m



5. (5 pts) When using a **vertical** differential element of area to solve for the location of $\bar{x}$ for the shaded area what expression would be correct for dA ?

- A. $(x^{0.667} + x^{1.5})/2 \, dx$
- B. $(x^{0.667} - x^{1.5}) \, dx$
- C. $(x^{0.667} + x^{1.5}) \, dx$
- D. $(x^{0.667}) \, dx$
- E. $(x^{1.5}) \, dx$

